

PRODUCT CASE STUDY

Aura Pulse90

An AI-Native Fitness Consistency System

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Product Builder · AI Product Management · aura-pulse90.web.app

4 Weeks

Build Time

Full-Stack PWA

Product Type

Live & Tested

Real Users

AI-Native Builder

Developer Profile

EXECUTIVE SUMMARY

Behavioral consistency by design

Most fitness products optimize for feature completeness. Aura Pulse 90 optimized for a single behavioral outcome: making consistency psychologically easier than churn.

Four structural bets drove this: a bounded 90-day program arc to reduce motivational fatigue, culturally contextual defaults to eliminate daily friction for South Asian users, AI coaching calibrated to the user's actual state, and an identity-reinforcing progression system that makes consistency feel visible.

- Structured 90-day transformation arc — reduces psychological fatigue of open-ended tracking
- 15-signal AI coaching pipeline — advice calibrated to recovery, streak, mood, sleep, and workout history
- South Asian-first diet defaults — eliminates configuration friction for the primary user segment
- Recovery-aware coaching — prevents overtraining through composite sleep + stress + intensity scoring
- Identity reinforcement system — XP, badges, and level titles create visible evidence of consistency

Shipped in 4 weeks using an AI-native development workflow. Every product decision — architecture, feature sequencing, tradeoffs, roadmap — was human-led.

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Why the existing market fails behavioral retention

Fitness app churn is not a motivation problem. It is a product design problem. Most apps are built around logging — they capture activity without sustaining behavior.

Gap	Mechanism of Failure	Design Response
Open-ended programs	No finish line creates sustained motivational load — users disengage when consistency breaks	90-day bounded program with re-engagement checkpoints at Days 14, 28, 56, 84
Paywalled coaching	Habit formation interrupted at 7–14 days when paywall activates — exactly when routines are forming	100% free with full AI coaching from day one
Western food defaults	Immediate friction for South Asian users — forces manual configuration of every meal	South Asian meal defaults loaded at onboarding; zero configuration required
Shallow streaks	Streak-only systems reward presence, not progress — no identity reinforcement	XP + 14 badges + 6 level titles tied to transformation milestones
Generic AI advice	Context-free coaching erodes trust — users disengage when advice doesn't reflect their actual state	15-signal context pipeline ensures advice is calibrated to user's real-time state

Sustainable transformation is a product architecture problem — not a willpower problem.

Primary persona — The Restarter

First-hand user. Research was continuous across multiple build iterations — direct observation, informal interviews, and behavioral signal from live app usage.

Attribute	Profile
Age	18–35
Location	Tier 2/3 Indian city — Guwahati, Indore, Patna, Coimbatore
Device	Android-first; limited data plan
Diet	South Asian — dal, roti, rice, paneer, oats, soya chunks
App history	Tried 2–4 fitness apps; quit after free trial expired or app felt irrelevant

Core frustration	Existing apps charge ₹1,000–2,000/mo and suggest foods they don't eat
Core motivation	Structured transformation with visible progress — not open-ended logging

Key research signals

- Diet module was the most-discussed feature across all user sessions — unprompted, consistent across backgrounds
- Users returned daily without push notification prompts — validated streak and dashboard design
- Feature requests emerged organically — food items, exercise variations, coach tone — confirming engagement depth
- Zero onboarding instruction needed for new users — strong signal of intuitive information architecture

03 — PRODUCT STRATEGY

The central bet

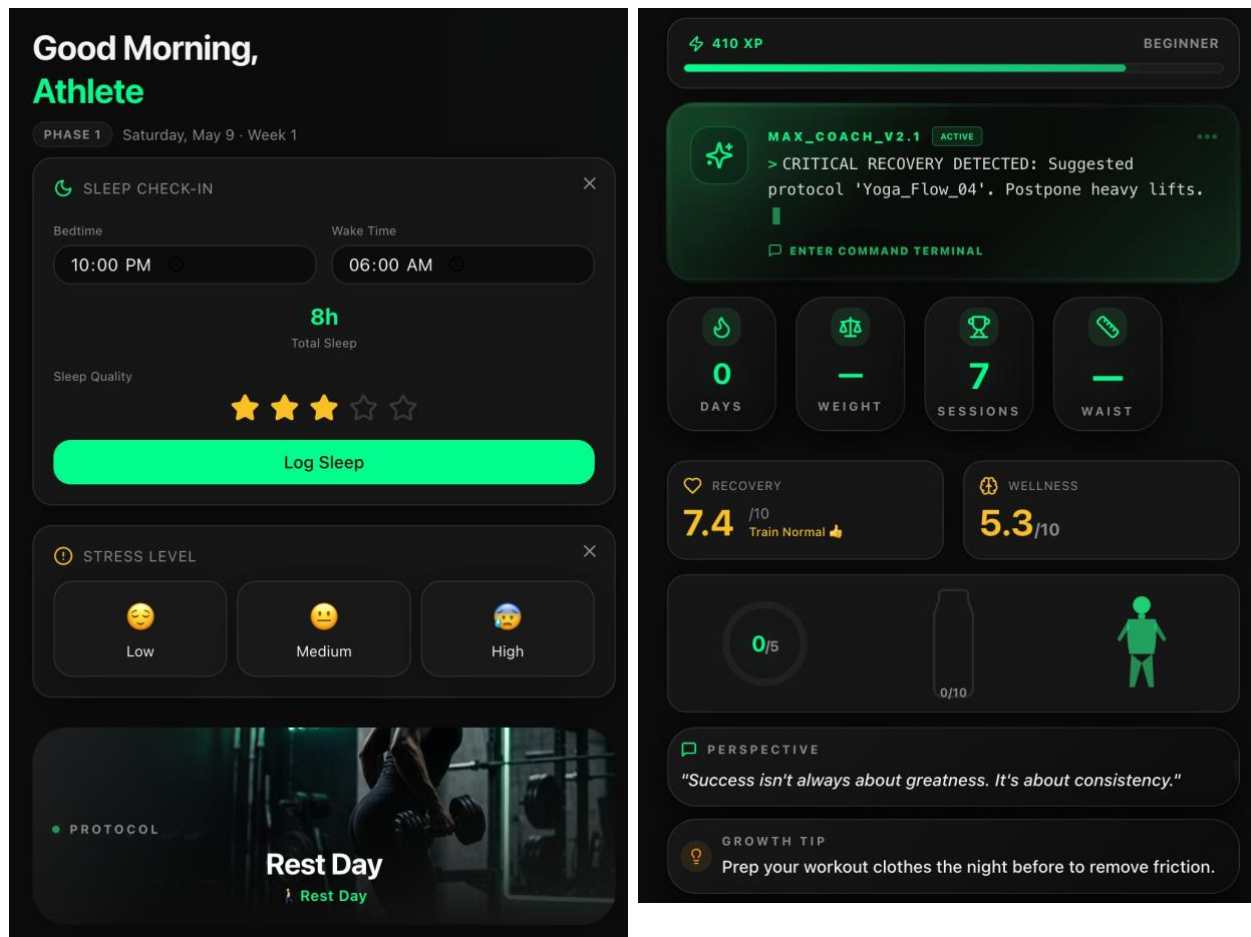
A bounded, AI-coached, culturally contextual fitness system — free, forever — could achieve stronger behavioral retention than subscription apps by eliminating the three primary friction points: cost, cultural irrelevance, and open-ended structure.

Competitive positioning

Dimension	Subscription Apps	Aura Pulse 90
Cost	₹500–1,000/mo after trial	100% Free
AI Coaching	Premium-only feature	Gemini AI — all users, from day one
Form Check	Not available / paid add-on	Built-in CV via TF.js MoveNet
Diet Defaults	Western / generic	South Asian — zero configuration
Program Structure	Open-ended	Structured 90-day arc
Progression	Basic streaks	XP, 14 badges, 6 level titles, leaderboard

The Daily Dashboard — single-screen motivation

The first screen a user sees every morning. Streak, XP progress, recovery score, mood check-in, and today's workout — zero decisions required to start.



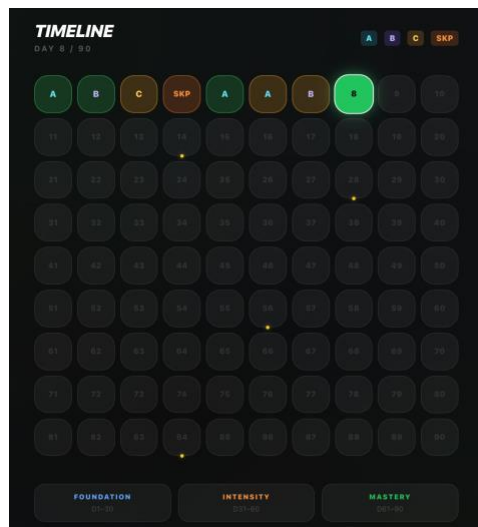
Daily Dashboard (sleep check-in view) · Dashboard (XP, recovery, Coach Max, today's stats) · Everything a user needs, one scroll

The 90-day structure — product rationale

- Bounded time horizons improve completion rates — validated by P90X, 75 Hard, and behavioral psychology research
- A visible finish line reduces the motivational load that causes open-ended app churn
- Natural milestone checkpoints (Day 14, 28, 56, 84) enable re-engagement before full churn occurs
- The program arc creates a coherent product narrative — users are on a journey, not managing a log

The 90-Day Timeline view

Every day is visible. Completed days show workout type (A/B/C), skipped days show SKP. Milestone checkpoints at Days 14, 28, 56, 84 are marked. Users can see exactly how far they've come and how far they have to go — the single most powerful anti-churn visual in the product.



90-Day Timeline · Day 8/90 shown · Foundation → Intensity → Mastery phases · Every day visible, every skip visible

04 — KEY PRODUCT DECISIONS

Five decisions that shaped the product

Decision 1 — Data Architecture: Local-First

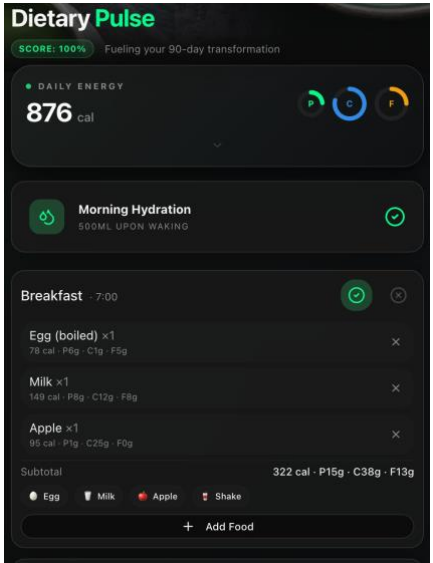
PROBLEM	Daily-use app requiring instant load times for users in low-connectivity Indian markets, with zero tolerance for per-request costs at free tier.
DECISION	localStorage as primary state + Firebase as async cloud backup, synced via debounced event bus.
WHY	Eliminates Firestore read costs at MVP scale. Sub-second load times with no network dependency for daily interactions. Offline-first for Tier 2/3 connectivity.
TRADEOFF	5MB localStorage ceiling limits power users with large photo logs. Accepted as a Phase 2 migration problem — storage upgrade triggered by scale, not day-one infrastructure spend.

Decision 2 — Authentication: Required, No Guest Mode

PROBLEM	AI coaching personalisation and cross-device data continuity require a persistent user identity. Leaderboard integrity requires authenticated participation.
DECISION	Sign-in required at onboarding via Google OAuth or Email/Password. No anonymous access path.
WHY	The product's core value — adaptive AI coaching calibrated to weeks of behavioral history — cannot be delivered without identity. A guest flow would deliver a materially weaker product.
TRADEOFF	Signup friction at activation. Accepted because identity gates the highest-value features. Users who complete onboarding receive a product that justifies the step.

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Decision 3 — Diet Module: South Asian—First Defaults



Dietary Pulse · Score: 100% · Macro rings (P/C/F) · Culturally familiar foods as defaults — Egg, Milk, Apple at breakfast · Zero configuration required

Decision 3 — Diet Module: South Asian—First Defaults	
PROBLEM	Global fitness apps default to Western food databases, creating immediate daily friction for South Asian users who must manually configure every meal.
DECISION	South Asian meal defaults loaded at first launch — contextually familiar foods, not Western substitutes. No configuration required.
WHY	Eliminates the single highest-friction point for the primary user segment. Positions the product as built for this market — not adapted for it. Creates a defensible differentiator requiring cultural knowledge, not just engineering.
TRADEOFF	Narrower initial TAM. Phase 2 requires expanded food database for diaspora and metro users with mixed eating patterns.

Decision 4 — 60+ Feature Flags: Experimentation Architecture	
PROBLEM	MVP scope must be controlled without removing features from the codebase. Future premium tier requires activation without a new release cycle.
DECISION	Every feature sits behind a boolean flag — toggled independently, per environment or per user group.
WHY	Enables instant rollback of any feature without deployment. Creates scaffolding for A/B testing. Future premium tier = enable flags for paying users, zero structural change.
TRADEOFF	Flag management complexity grows at scale. Requires a flag registry and audit process before significant user base expansion.

PM-led execution using LLM co-engineering

Aura Pulse 90 was built using a fully AI-native development workflow — not as a workaround, but as a deliberate model of how PM-led product development works in 2026.

Tool	Role in Build	What It Replaced
Claude	Architecture reasoning, PRD structuring, tradeoff analysis	Senior engineer consultation
Cursor / Copilot	In-IDE code completion and debugging	Junior engineering pair
Lovable	Rapid UI prototyping from product specs	Frontend design handoff
Gemini AI	AI coaching pipeline design and testing	ML team scoping sessions
ChatGPT	Implementation debugging, API integration	Stack Overflow + documentation sprint

What it feels like to use Aura Pulse 90

A day in the product — from morning check-in to workout completion. Each interaction is designed to reduce a specific friction point in the behavioral consistency loop.

Morning — The Daily Check-in

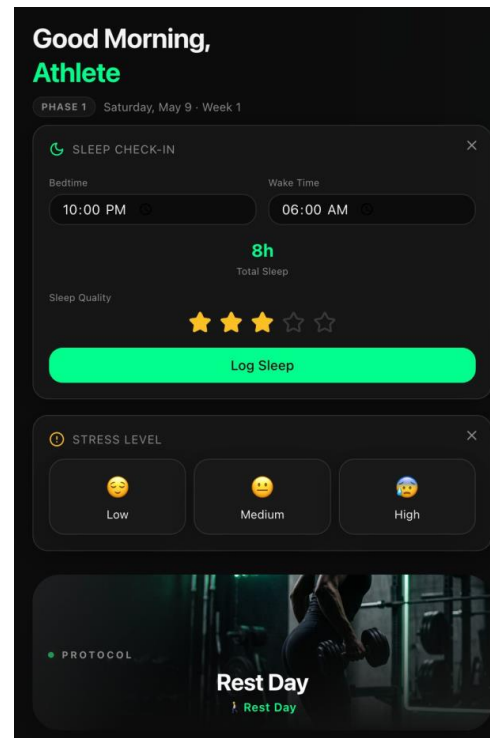
The app opens to a personalised greeting. Before anything else, the user logs two signals that feed directly into today's AI coaching:

- 1

Sleep Check-in
Bedtime, wake time, and quality (1–5 stars). Takes 15 seconds. Feeds directly into the recovery score and shapes Coach Max's morning message — a user with 5 hours of poor sleep gets a different coaching tone than one with 8 hours of deep sleep.
- 2

Stress Level
One tap: Low / Medium / High. The second input to the composite recovery score. High stress → coach adjusts to a lighter session or suggests a breathing protocol instead.
- 3

Recovery Score Displayed
The composite score (sleep quality + hours + stress + previous workout intensity) appears immediately. A score of 7.4/10 = 'Train Normal.' A score under 5 = coach proactively suggests rest or active recovery. Users quickly learn that honest inputs produce better recommendations.



Left: Dashboard showing Coach Max alert, recovery score 7.4/10, wellness 5.3/10 ·
 Right: Sleep check-in and stress level logging — the two signals that personalise every coaching session

Planning — The Workout Session

After check-in, the user sees today's protocol. The product removes the decision of what to do — it's already decided based on the weekly schedule and program phase.

4 Session Protocol View

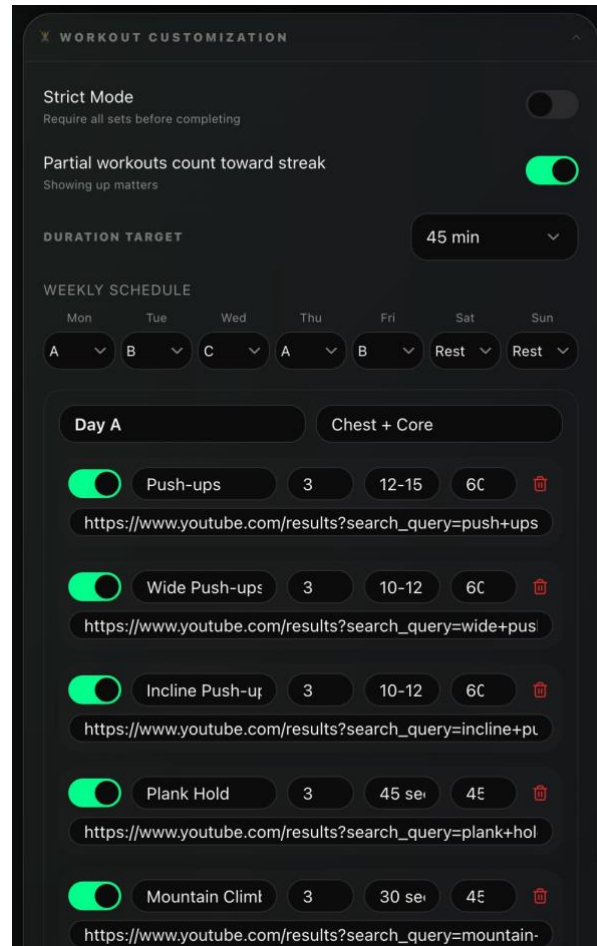
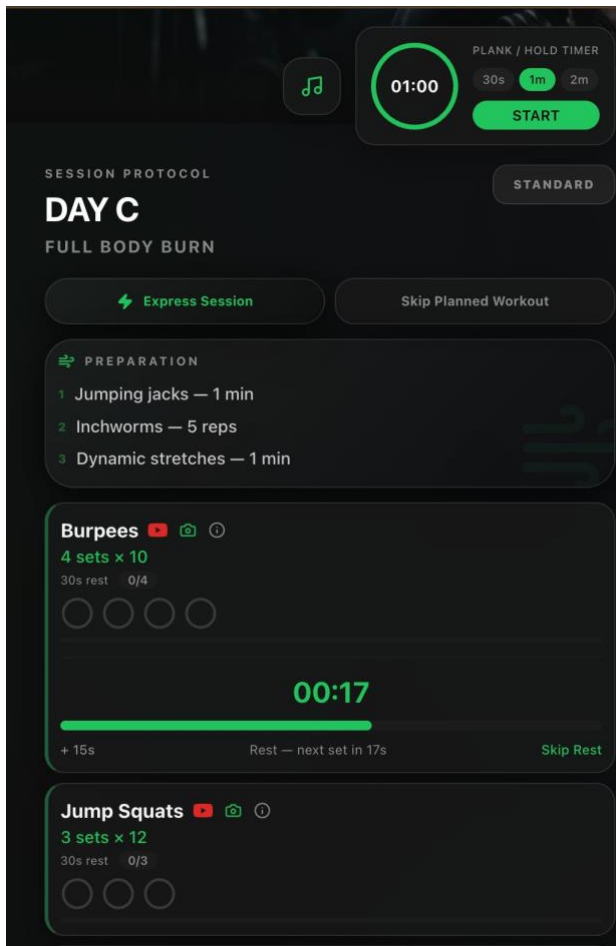
Day C — Full Body Burn. Shows preparation exercises, the full workout (Burpees 4×10, Jump Squats 3×12), rest timers, and a Plank/Hold timer. A music button opens Spotify or YouTube. Express Session and Skip Planned Workout are both visible — the product respects user autonomy without hiding the full session.

5 Rest Timer

Between sets, a countdown timer runs automatically. Users can add +15s or tap Skip Rest. This single interaction — visible, controllable rest timing — removes the most common source of workout distraction (checking the phone for time).

6 Workout Customization

The weekly schedule (A/B/C/Rest) is fully editable. Strict Mode, duration targets, and per-exercise toggles give users control without overwhelming them on day one — all customisation is accessed through a secondary screen, keeping the primary workout view clean.



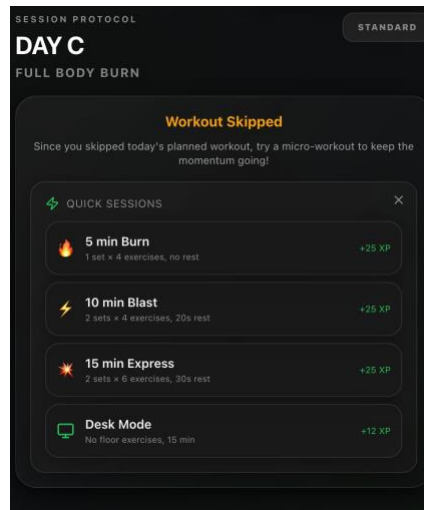
Left: Active workout — Day C Full Body Burn, rest timer at 00:17, set completion circles, Express Session option
 Right: Workout customization — weekly schedule, duration target, per-exercise toggles with YouTube links

When Life Intervenes — The Skip Flow

The single most important interaction in any habit-formation product is what happens when a user misses a session. Most apps do nothing. Aura Pulse 90 treats a skip as a design moment.

7 Workout Skipped Screen

When a user marks a workout as skipped, the app immediately surfaces Quick Sessions: 5 Min Burn (+25 XP), 10 Min Blast (+25 XP), 15 Min Express (+25 XP), Desk Mode (+12 XP — no floor exercises). The message: 'Since you skipped today's planned workout, try a micro-workout to keep the momentum going.' Showing up in any form counts. Partial workouts count toward streak (configurable).



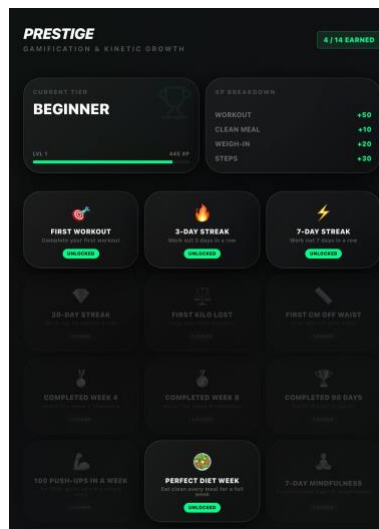
Workout Skipped — Day C · Quick Sessions offered immediately: 5 Min Burn, 10 Min Blast, 15 Min Express, Desk Mode
· Every skip becomes a re-engagement moment, not a dead end

Progress — Gamification & Identity

The PRESTIGE screen is the product's identity layer. It answers the question every user asks after week two: 'Am I actually getting anywhere?'

8 PRESTIGE — Gamification & Kinetic Growth

Current tier (BEGINNER → WARRIOR → ELITE → LEGEND), XP breakdown by action (Workout +50, Clean Meal +10, Weigh-in +20, Steps +30), and 14 badges across fitness milestones. 4/14 earned. Locked badges show what's next — First Kilo Lost, 30-Day Streak, Completed Week 4. The locked badges are as motivating as the unlocked ones: they make the next milestone visible.



PRESTIGE — Gamification & Kinetic Growth · 4/14 badges earned · Beginner tier, 445 XP · XP breakdown by action type · Locked milestones create forward pull

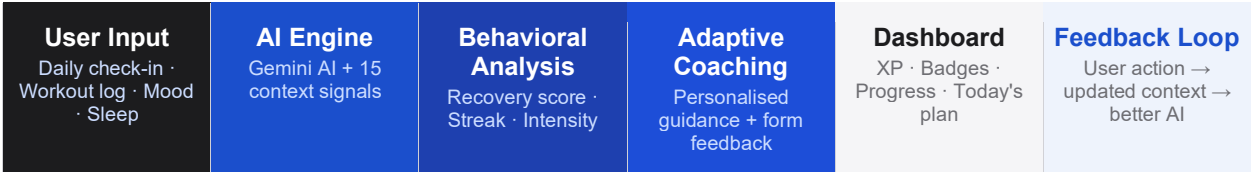
Each feature as a behavioral system

Features were not built as utilities. Each module addresses a specific failure point in the retention loop.

Feature	Behavioral Problem Addressed	Design Mechanism
Daily Dashboard	Friction at task initiation — users don't know where to start	Single-screen state: streak, XP, mood, today's workout — reduces decision load to zero
AI Coach (Max)	Generic advice erodes trust and disengagement accelerates	15-signal context pipeline — advice reflects user's real recovery, history, and progression state
Recovery Score	Overtraining causes injury and dropout; rest feels like failure	Composite score (sleep + stress + intensity) makes recovery a measurable, rewarded action
Gamification System	Consistency without visible progress feels like stagnation	XP + 14 badges + 6 level titles create identity markers at every milestone
Skip → Quick Session	A skipped day becomes a churn event in most apps	Immediate micro-workout offer turns every skip into a re-engagement opportunity
South Asian Diet	Daily configuration friction compounds into abandonment	Cultural defaults eliminate meal-logging friction from day one — no setup required
Form Check (TF.js)	Unsafe solo training causes injury and erodes confidence	Real-time joint angle detection + voice feedback — removes need for trainer presence
PWA Architecture	App Store barrier reduces activation rate	Installable from browser — eliminates store approval, update cycle, and download friction
60+ Feature Flags	New features risk breaking existing behavioral loops	Independent flag control — any feature can be disabled instantly without deployment

System architecture

The product is a closed behavioral feedback loop — every user action refines the AI context, improving subsequent coaching quality.



Closed feedback loop — each user action refines subsequent AI coaching

What the MVP validated

No analytics instrumentation in v1.0 — a known gap. Signals are behavioral and qualitative, observed across multiple live user sessions.

Hypothesis	Signal Observed	Status
Streak mechanic drives daily return	Users returned daily without prompts across all sessions	Validated
South Asian defaults reduce friction	Diet module was most-discussed feature — unprompted, consistent	Strongly validated
AI coaching feels contextual	Users referenced specific coach responses in follow-up sessions	Validated
90-day structure improves engagement	No users described the program as 'overwhelming' — open-ended framing did	Validated
App usable without instructions	Zero onboarding support required across all test users	Validated
Analytics gap	No funnel data — activation rate, drop-off, session length unmeasured	Gap — Phase 2 P0

What gets measured at scale

Metric	Target & Reasoning
D7 Retention	≥40% — primary signal for behavioral loop design quality
90-day Completion	≥15% — validates core product hypothesis on bounded programs
First Workout Conversion	≥65% — measures onboarding friction and activation design
Streak Recovery Rate	≥30% within 3 days — measures re-engagement system effectiveness
Coach Sessions / Week	≥3 per active user — validates AI coaching as a retention driver
Diet Module Usage	≥80% of DAU — validates cultural positioning and reduces churn risk

What changed, what didn't, what I'd do differently

What held

- Strongest product signal in the build — validated cultural positioning as a real differentiator, not a cosmetic feature — South Asian diet defaults
- Users consistently described it as 'more manageable' than open-ended apps — structural hypothesis confirmed — 90-day bounded structure
- The re-engagement moment after a missed day reduced guilt and kept users in the product — an underinvested pattern in most fitness apps — Skip → Quick Session flow
- Compressed 4–6 months of traditional build into 4 weeks without reducing product decision quality — AI-native development

What I'd change

- No funnel data meant decisions relied on qualitative signal only. Next build: PostHog live before first user — Analytics from day one
- 10 feature areas in an MVP is wide. A tighter core loop — workout + coach + streak — would have produced cleaner retention signals faster — Narrower initial scope
- Feedback was continuous but unstructured. A weekly 30-minute session format would have surfaced insights more reliably — Structured interview cadence

Product management is the discipline of making better decisions than the available information justifies — then updating fast when new signals arrive.

Stability → Scale → Growth

Phase 1 — Stability (Weeks 1–2)	Phase 2 — Scale (Month 2)	Phase 3 — Growth (Month 3+)
Re-enable auth guard on coachAction — P0 cost risk	PostHog analytics — blocks data-driven decisions	Gemini 2.0 upgrade with streaming responses
Migrate photos to Firebase Storage — removes localStorage ceiling	Migrate primary state to Firestore — eliminates 5MB cap	Native mobile via Capacitor — App Store + Play Store
Remove Supabase legacy config — security hygiene	Decompose Dashboard + WorkoutPage monoliths	Social milestone sharing — organic acquisition driver
Add per-page error boundaries — prevents full app crash	Unit tests for core algorithms (recovery, streak, diet score)	Premium tier — ₹299–₹799/mo advanced AI features
	Re-engagement flow design — addresses highest churn moment	B2B white-label — ₹5,000–₹20,000/mo per gym/coach partner

Competencies evidenced by this project

Prioritisation Triaged 60+ features and technical debt items across four build phases using impact/effort reasoning. Auth risk → P0. Analytics → Phase 2. Scope reduction → deliberate.	Systems Thinking Designed 20+ TypeScript interfaces, a local-first sync architecture, and a closed AI feedback loop. Every component connects to the behavioral outcome.
Behavioral UX Each feature designed to reduce friction at a specific point in the retention loop — not as a utility, but as a behavioral intervention. The skip flow is the clearest example.	AI-Native Execution Shipped a full-stack AI product in 4 weeks with no engineering background — using Claude, Cursor, Lovable, and Gemini as co-engineers under PM direction.
Roadmap Planning Three-phase roadmap with explicit prioritisation rationale across Stability, Scale, and Growth — including deferred decisions and their triggers.	Product Strategy Positioned against a crowded market by owning an underserved segment (South Asian, Tier 2/3) rather than competing on feature parity with well-funded incumbents.
Rapid Experimentation 60+ feature flags enabled weekly experimentation cycles without deployment risk. Architecture designed for testing from day one.	Tradeoff Reasoning Five major architecture decisions documented with Problem / Decision / Why / Tradeoff structure — showing conscious choice over convenient default.

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